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ecologically fragile region, had been set aside by authorities a decade back.

In 2013, the National Remote Sensing Centre published a study that showed that the Lhonak lake has a high outburst value of 42 per cent and that, according to satellite images, between 1962 and 2008, the South Lhonak glacier retreated 1,941 metres, making the lake one of the fastest growing in the Sikkim Himalayas.

Yet another study in 2019, by scientists Ashim Sattar and Ajanta Goswami of the Indian Institute of Technology, Roorkee, and Anil V. Kulkarni of the Indian Institute of Science in Bengaluru, warned of an impending GLOF calamity involving the Lhonak lake. "At Chungthang town, 62.35 kilometres from South Lhonak lake, the flood wave would potentially inundate settlements along the bank of the flow channel...," the study predicted. According to a press note issued by the National Disaster Management Authority (NDMA) on October 4, the Himalayas are host to about 7,500 lakes and estimates made through remote sensing techniques suggest 10 per cent of these are in Sikkim, of which 25 are assessed to be at high risk.

"The rapid retreat of glaciers in the Himalayas and their impact is alarming. There are about 25 lakes in Sikkim that are at risk of flooding," says Dr Gourab Bhattacharya, assistant professor in the Department of Earth Sciences at Tennessee Tech University.

Such has been the devastation left behind, especially in Mangan, Gangtok, Pakyong and Namchi districts, that the Sikkim government is yet to fully fathom the magnitude of its impact. Vijay Bhushan Pathak, chief secretary, Sikkim, tells INDIA TODAY that the government was still carrying out relief and evacuation. According to Pathak, 37 lives have been lost so far and 81 people are still missing, including several army personnel. "Around 2,000 houses and 922 shops have been damaged; most have been destroyed fully. A hundred villages across many panchayat areas have been hit," he adds. In all, about 90,000 people have been affected.

A source in the state government says that it would take at least another fortnight to clear the accumulated debris—five feet deep in some places—left behind by the deluge, since many streets are too narrow for large machines to enter. He adds that the Swiss Agency for Development and Cooperation and the NDMA have been engaged in assessing the risks of proglacial lakes in Sikkim and had in fact released guidelines in 2020 to manage precisely such GLOF threats. The guidelines had proposed the framing of a single legislation to control building activity in this eco-sensitive region. Though the Sikkim State Disaster Management Authority is working with the two organisations, a foolproof early warning system is yet to be put in place.

THE RAPIDLY EXPANDING LHONAK LAKE HAD LONG BEEN RED-FLAGGED AS BEING RIPE FOR A 'GLOF' EVENT

"The Central Water Commission and the state's water resources department are actively assessing the risks and looking for ways to mitigate the loss downstream," the source says.

The calamity has also raised questions on the viability of hydel power projects and their attendant dams that are being built along the Teesta—one of the most-dammed rivers in India. With the Sikkim Urja or the Teesta III dam—pitched as Sikkim's biggest—washed away and the dams for the Teesta V and VI hydropower projects damaged, demands for an expert reassessment of their strength and viability are getting stronger. "Flood engineering or disaster management experts must be roped in to inspect or maintain the dams. Naturally, they are built to withhold floods, but they have to be strong enough to endure massive flash floods such as this," Dr Bhattacharya says.

The National Green Tribunal (NGT) has now issued notices to the Sikkim government, Sikkim Urja Ltd and the National Hydroelectric Power Company (NHPC) Ltd for a hearing related to the breach of the Teesta-III dam. In 2014, local Lepcha bodies had protested the environmental clearance granted to the (proposed) Teesta IV project, but after the NHPC protested that it was at a safe distance from the threat of GLOFs, the NGT gave a go-ahead in 2017. In 2006, too, a similar objection to the Teesta III project before the National Environment Appellate Authority (NEAA) was made in vain. The state's latest flood of misery should not surprise anyone then. ■